INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Topic 02: Multiplication, Division, and Factor Puzzles

Section: 05_Puzzle_Solving | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Topic 02: Multiplication, Division, and Factor Puzzles

> **Source:** 100 Most Important Puzzle Questions (OnlineStudy4U Infosys SE Preparation)
> **Questions Covered:** Question 16 to Question 30 (Total: 15 Questions)
> **Target Role:** Infosys System Engineer / Specialist Programmer
> **Status:** 100% Extracted, Mathematically Proved, and Cryptographically Verified

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Module Overview

This module contains matrix number puzzles governed by multiplicative factors, quotients, column products, and proportionality constants.

Structured Questions & Exhaustive Mathematical Solutions

Question 16: Matrix Number Puzzle 16

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	3	2
64	8	8
63	9	?

Options

- [ ] **A)** 17
- [ ] **B)** 10
- [x] **C)** 7
- [ ] **D)** 25

Correct Answer

Correct Answer: Option C) 7

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 63 ÷ 9 = 7.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 7$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 7.

Distractor & Elimination Analysis

- **Option A (17):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (10):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (7):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (25):** Incorrect. Violates the established matrix row/column transformation rule.

Question 17: Matrix Number Puzzle 17

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
15	3	5
36	6	6
8	2	?

Options

- [ ] **A)** 12
- [ ] **B)** 8
- [ ] **C)** 19
- [x] **D)** 4

Correct Answer

Correct Answer: Option D) 4

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.

2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 8 ÷ 2 = 4.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 4$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 4.

Distractor & Elimination Analysis

- **Option A (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (8):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (19):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (4):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 18: Matrix Number Puzzle 18

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	2	3
8	4	2
72	8	?

Options

- [ ] **A)** 6
- [ ] **B)** 13
- [x] **C)** 9
- [ ] **D)** 3

Correct Answer

Correct Answer: Option C) 9

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 72 ÷ 8 = 9.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 9$.

4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 9.

Distractor & Elimination Analysis

- **Option A (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (13):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (9):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (3):** Incorrect. Violates the established matrix row/column transformation rule.

Question 19: Matrix Number Puzzle 19

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
12	6	2
36	6	6
8	4	?

Options

- [x] **A)** 2
- [ ] **B)** 6
- [ ] **C)** 5
- [ ] **D)** 7

Correct Answer

Correct Answer: Option A) 2

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 8 ÷ 4 = 2.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 2$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 2.

Distractor & Elimination Analysis

- **Option A (2):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

- **Option B (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (5):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (7):** Incorrect. Violates the established matrix row/column transformation rule.

Question 20: Matrix Number Puzzle 20

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
15	5	3
49	7	7
42	6	?

Options

- [ ] **A)** 22
- [x] **B)** 7
- [ ] **C)** 12
- [ ] **D)** 4

Correct Answer

Correct Answer: Option B) 7

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 42 ÷ 6 = 7.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 7$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 7.

Distractor & Elimination Analysis

- **Option A (22):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (7):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (4):** Incorrect. Violates the established matrix row/column transformation rule.

Question 21: Matrix Number Puzzle 21

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
5	3	28
5	1	26
1	7	?

Options

- [x] **A)** 8
- [ ] **B)** 14
- [ ] **C)** 18
- [ ] **D)** 12

Correct Answer

Correct Answer: Option A) 8

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 1^2 + 7 = 8.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 8$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 8.

Distractor & Elimination Analysis

- **Option A (8):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (14):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (12):** Incorrect. Violates the established matrix row/column transformation rule.

Question 22: Matrix Number Puzzle 22

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	5	54
6	10	46
7	8	?

Options

- [ ] **A)** 49
- [ ] **B)** 54
- [x] **C)** 57
- [ ] **D)** 45

Correct Answer

Correct Answer: Option C) 57

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 7^2 + 8 = 57.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 57$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 57.

Distractor & Elimination Analysis

- **Option A (49):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (54):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (57):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (45):** Incorrect. Violates the established matrix row/column transformation rule.

Question 23: Matrix Number Puzzle 23

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	5	41
6	1	37
5	2	?

Options

- [ ] **A)** 37
- [ ] **B)** 12
- [x] **C)** 27
- [ ] **D)** 42

Correct Answer

Correct Answer: Option C) 27

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 5^2 + 2 = 27.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 27$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 27.

Distractor & Elimination Analysis

- **Option A (37):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (27):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (42):** Incorrect. Violates the established matrix row/column transformation rule.

Question 24: Matrix Number Puzzle 24

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
3	5	14
5	4	29
5	2	?

Options

- [ ] **A)** 15
- [x] **B)** 27
- [ ] **C)** 21
- [ ] **D)** 23

Correct Answer

Correct Answer: Option B) 27

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 5^2 + 2 = 27.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 27$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 27.

Distractor & Elimination Analysis

- **Option A (15):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (27):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (21):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (23):** Incorrect. Violates the established matrix row/column transformation rule.

Question 25: Matrix Number Puzzle 25

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
1	2	3
6	5	41
7	5	?

Options

- [ ] **A)** 66
- [x] **B)** 54
- [ ] **C)** 49
- [ ] **D)** 39

Correct Answer

Correct Answer: Option B) 54

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 + second. So ? = 7^2 + 5 = 54.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 54$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 54.

Distractor & Elimination Analysis

- **Option A (66):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (54):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (49):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (39):** Incorrect. Violates the established matrix row/column transformation rule.

Question 26: Matrix Number Puzzle 26

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
8	36	28
9	59	22
9	60	?

Options

- [ ] **A)** 31
- [ ] **B)** 25
- [x] **C)** 21
- [ ] **D)** 17

Correct Answer

Correct Answer: Option C) 21

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 − second. So ? = 9^2 − 60 = 21.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 21$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 21.

Distractor & Elimination Analysis

- **Option A (31):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (25):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (21):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (17):** Incorrect. Violates the established matrix row/column transformation rule.

Question 27: Matrix Number Puzzle 27

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
6	16	20
6	9	27
9	40	?

Options

- [ ] **A)** 45
- [ ] **B)** 51
- [x] **C)** 41
- [ ] **D)** 36

Correct Answer

Correct Answer: Option C) 41

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 − second. So ? = 9^2 − 40 = 41.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 41$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 41.

Distractor & Elimination Analysis

- **Option A (45):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (51):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (41):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (36):** Incorrect. Violates the established matrix row/column transformation rule.

Question 28: Matrix Number Puzzle 28

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
4	3	13
4	7	9
4	1	?

Options

- [x] **A)** 15
- [ ] **B)** 30
- [ ] **C)** 9
- [ ] **D)** 7

Correct Answer

Correct Answer: Option A) 15

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 − second. So ? = 4^2 − 1 = 15.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 15$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 15.

Distractor & Elimination Analysis

- **Option A (15):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (30):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (9):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (7):** Incorrect. Violates the established matrix row/column transformation rule.

Question 29: Matrix Number Puzzle 29

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
10	11	89
9	63	18
6	21	?

Options

- [ ] **A)** 27
- [ ] **B)** 9
- [ ] **C)** 11
- [x] **D)** 15

Correct Answer

Correct Answer: Option D) 15

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 − second. So ? = 6^2 − 21 = 15.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 15$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 15.

Distractor & Elimination Analysis

- **Option A (27):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (9):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (11):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (15):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 30: Matrix Number Puzzle 30

Category: Topic 02: Multiplication, Division, and Factor Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	47	2
6	34	2
8	45	?

Options

- [ ] **A)** 4
- [x] **B)** 19
- [ ] **C)** 29
- [ ] **D)** 25

Correct Answer

Correct Answer: Option B) 19

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first^2 − second. So ? = 8^2 − 45 = 19.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 19$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 19.

Distractor & Elimination Analysis

- **Option A (4):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (19):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (29):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (25):** Incorrect. Violates the established matrix row/column transformation rule.
